

Multi-Subnet DHCP Configuration & DHCP Relay Agent Implementation

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1. Executive Summary & Lab Topology

This lab demonstrates the deployment of centralized dynamic IP addressing across a multi-subnet routing environment using Cisco IOS. A single router (**R2**) acts as the centralized DHCP server hosting multiple address pools, while a secondary infrastructure router (**R1**) acts as both a DHCP client and a **DHCP Relay Agent** (using the `ip helper-address` mechanism) to bridge broadcast domains.

Network Addressing Plan

Subnet Purpose	Network Address	Gateway IP	DHCP Role	Excluded Range
PC2 LAN (Left)	192.168.2.0 /24	192.168.2.1 (R2 G0/1)	Direct Server DHCP	192.168.2.1 – 192.168.2.10
WAN Link (Middle)	203.0.113.0 /30	203.0.113.1 (R2 G0/0)	Direct Server DHCP	203.0.113.1
PC1 LAN (Right)	192.168.1.0 /24	192.168.1.1 (R1 G0/1)	Relayed via R1	192.168.1.1 – 192.168.1.10

2. Device Roles & Configuration Commands

2.1 Centralized DHCP Server (R2 Configuration)

Router **R2** is the core DHCP manager. It defines the excluded management addresses and runs three individual pools. During configuration, standard Cisco IOS syntax was strictly followed, resolving initial subnet mask formatting errors.

```
R2> enable
R2# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.10
R2(config)# ip dhcp excluded-address 192.168.2.1 192.168.2.10
R2(config)# ip dhcp excluded-address 203.0.113.1

R2(config)# ip dhcp pool pool1
R2(dhcp-config)# net 192.168.1.0 /24
% Invalid input detected at '^' marker.
R2(dhcp-config)# network 192.168.1.0 255.255.255.0
R2(dhcp-config)# dns-server 8.8.8.8
R2(dhcp-config)# domain-name Momolab.com
R2(dhcp-config)# default-router 192.168.1.1

R2(dhcp-config)# ip dhcp pool pool12
R2(dhcp-config)# network 192.168.2.0 255.255.255.0
R2(dhcp-config)# dns-server 8.8.8.8
R2(dhcp-config)# domain-name Momolab.com
R2(dhcp-config)# default-router 192.168.2.1

R2(dhcp-config)# ip dhcp pool pool3
R2(dhcp-config)# network 203.0.113.0 255.255.255.252
```

2.2 DHCP Client & Relay Agent (R1 Configuration)

Router **R1** serves a dual purpose. Its upstream interface (GigabitEthernet0/0) functions as a **DHCP Client** to dynamically receive its WAN link IP from R2. Its downstream interface (GigabitEthernet0/1) serves as a **DHCP Relay Agent** to forward local broadcast discover messages as unicast requests to R2.

```

R1> enable
R1# configure terminal

R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip address dhcp
R1(config-if)# no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0 assigned DHCP address 203.0.113.2,
mask 255.255.255.252, hostname R1

R1(config-if)# interface GigabitEthernet0/1
R1(config-if)# ip helper-address 203.0.113.1

```

3. Verification Results & Client Output

3.1 WAN Link Address Assignment

As shown in the R1 CLI syslog output, the interface GigabitEthernet0/0 successfully initiated a DHCP handshake over the point-to-point link. It was assigned the first available unexcluded address from **pool3**:

Assigned IP: 203.0.113.2 /30

3.2 PC2 Verification (Direct DHCP Subnet)

PC2 resides inside the 192.168.2.0/24 subnet directly attached to the DHCP server infrastructure. Triggering a renewal fetched the first valid lease past the excluded boundary (.1 to .10):

```

C:\> ipconfig /renew

IP Address. . . . . : 192.168.2.11
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.2.1
DNS Server. . . . . : 8.8.8.8

C:\> ipconfig /all
Connection-specific DNS Suffix. . : Momolab.com
Physical Address. . . . . : 00E0.B087.76E5

```

3.3 PC1 Verification (Relayed DHCP Subnet)

PC1 resides behind the Relay Agent (R1). The initial request timed out because the relay agent link was initializing. The subsequent request successfully traversed R1 via the `ip helper-address` policy, obtaining its lease parameters from R2's **pool1**:

```
C:\> ipconfig /renew
DHCP request failed.

C:\> ipconfig /renew
IP Address. . . . . : 192.168.1.12
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DNS Server. . . . . : 8.8.8.8
DHCP Server . . . . . : 203.0.113.1
Connection-specific DNS Suffix. . : Momolab.com
```

Key Observation & Analysis

Notice that PC1 correctly identifies the DHCP Server as **203.0.113.1** (R2), even though its default gateway is **192.168.1.1** (R1). This proves that the Relay Agent successfully injected its own interface IP into the DHCP Giaddr field, enabling R2 to select the correct pool (pool1) and unicast the response back across the WAN link.

4. Conclusion & Technical Insights

1. **Syntax Accuracy:** In Cisco IOS, standard network pools must define the mask explicitly using dotted-decimal notation (e.g., 255.255.255.0) rather than CIDR slash notation (/24), as caught by the syntax checking engine during the R2 configuration.
2. **Relay Efficiency:** The implementation of `ip helper-address` eliminates the need to maintain a dedicated DHCP server within every distinct broadcast domain, centralizing network administration and security monitoring.